



GROUP 3

EURO-PARK

From Reactive Queues to Predictive Flow Management
Leveraging Waiting Time Forecasts to Improve Park Performance

Objectives

The objective of this mission is to transform historical operational data into predictive and actionable insights that improve both visitor experience and park performance.

I. Forecast Waiting Time

- Predict waiting times by attraction and hour in the short term
- Anticipate congestion before queues form
- Provide reliable, forward-looking visibility for operations teams

II. Improve KPIs

- Proactively manage visitor flows and reduce average waiting times
- Support operational decisions such as staffing, routing, and capacity allocation
- Enable strategic planning through attendance forecasts for pricing, maintenance, and resource planning

Ultimately, the goal is to move from reactive queue management to data-driven, proactive decision-making, improving key KPIs such as waiting time, visitor satisfaction, and operational efficiency.

Problem Statement

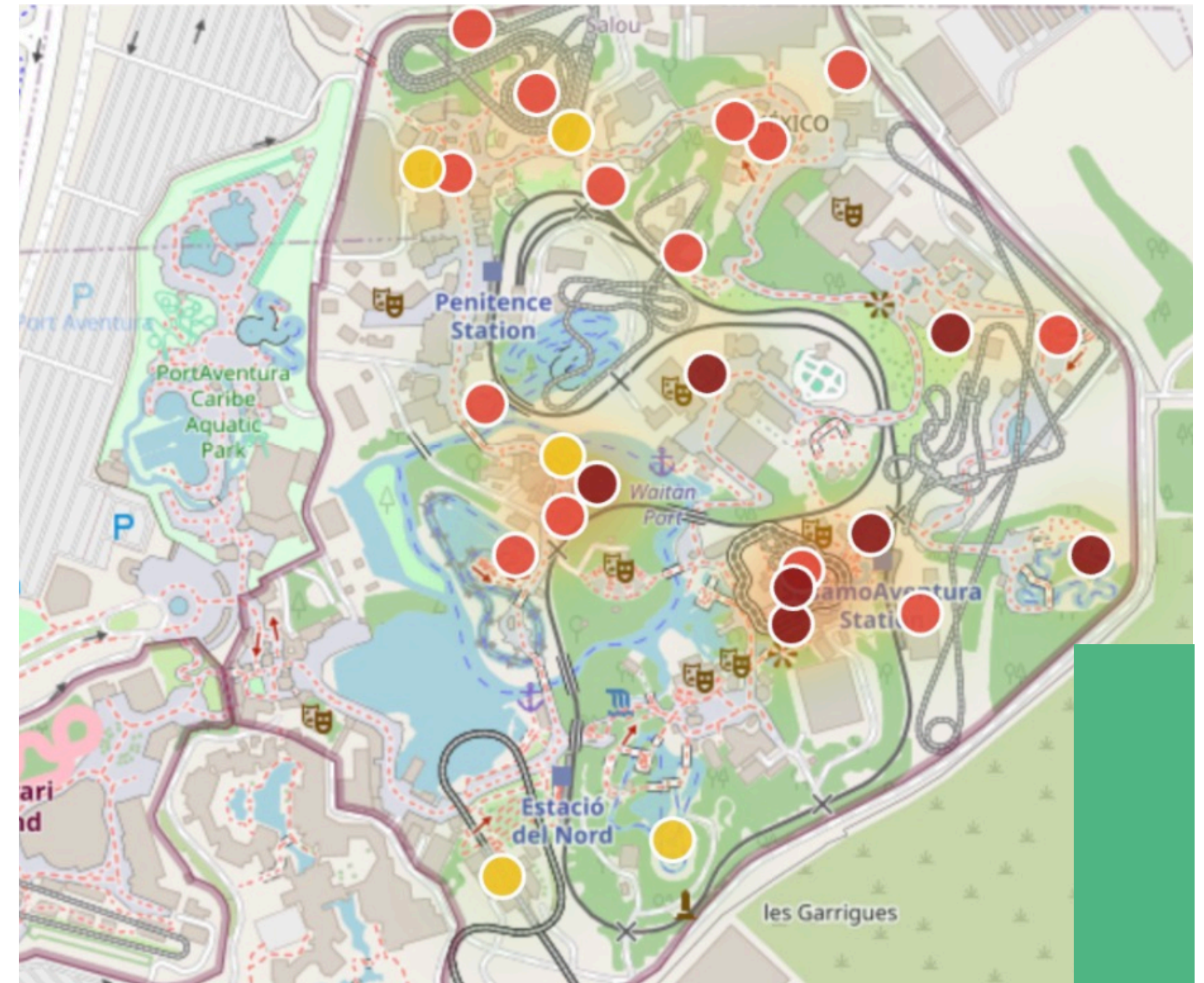


CONTEXT

- Euro-park's visitor experience is strongly impacted by waiting times
- Long unpredictable queues lead to reduced visitor satisfaction
- Large amounts of operational data are available, but they are mostly used retrospectively

COMPLICATIONS

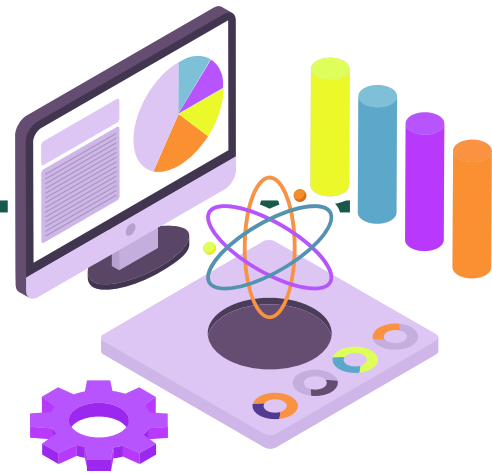
- Predicting waiting times is complex
- Today, teams lack forward-looking visibility, leading to reactive queue management and limiting the ability to plan effectively.



Problem Statement

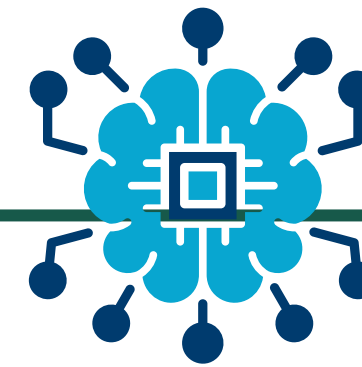
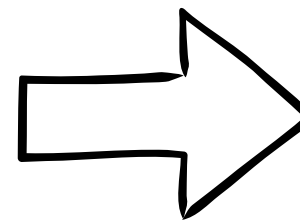


Client wants to assess the use of Forecasting Models on improving customer satisfaction with additional operational benefits



Forecasting the waiting time for rides across the week

- Day of visit plan
- Optimal route plan
- Crowd management



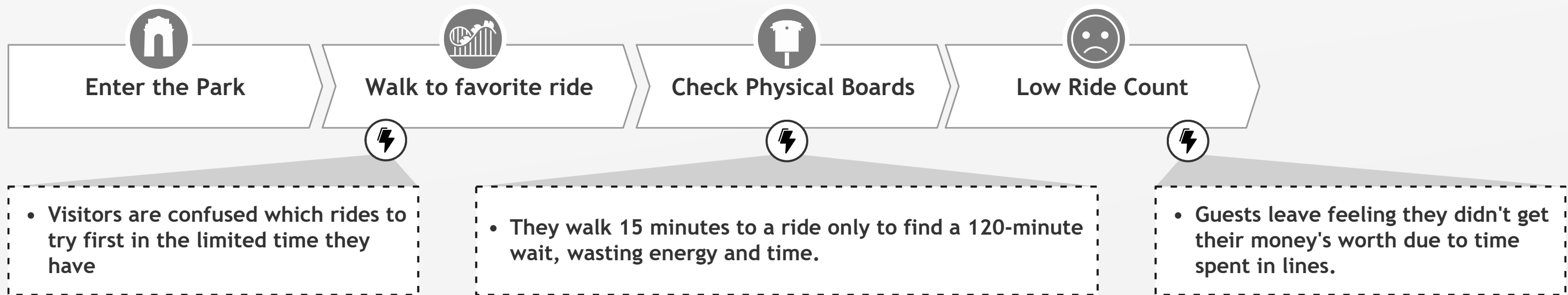
Revenue Forecasting
Revenue Optimization
Staffing management
Maintenance intervals

Challenge: Congestion varies throughout the day depending on many factors even on days with similar visitor numbers

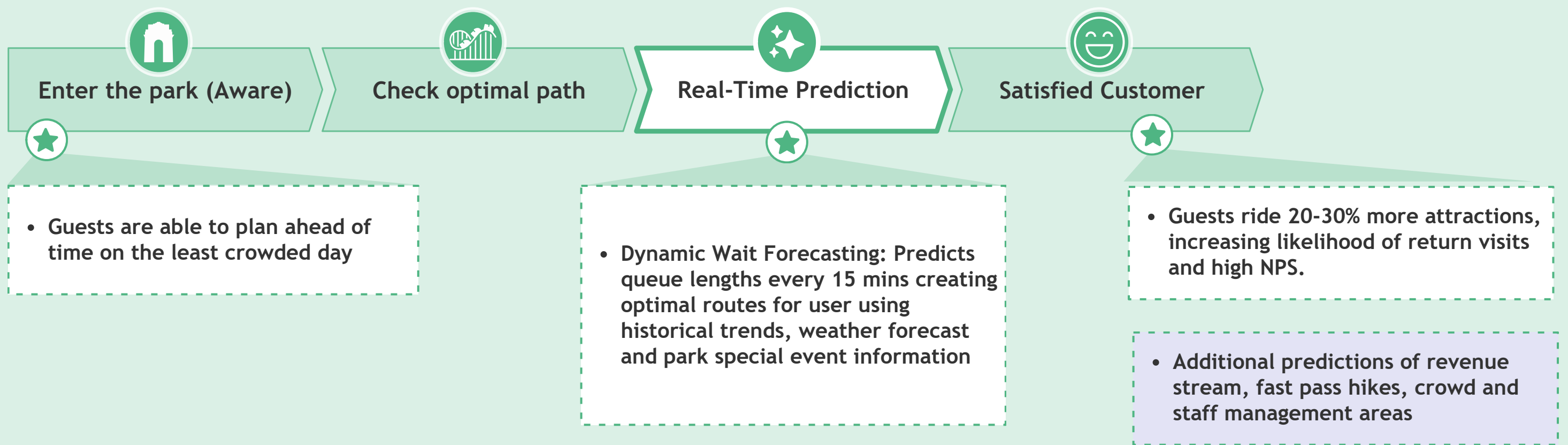
Enhanced Customer Flow

Our forecasting model doesn't just inform guests - it manages park flow. Reduces operational strain and creates a premium experience for every visitor.

Current process



Augmented process

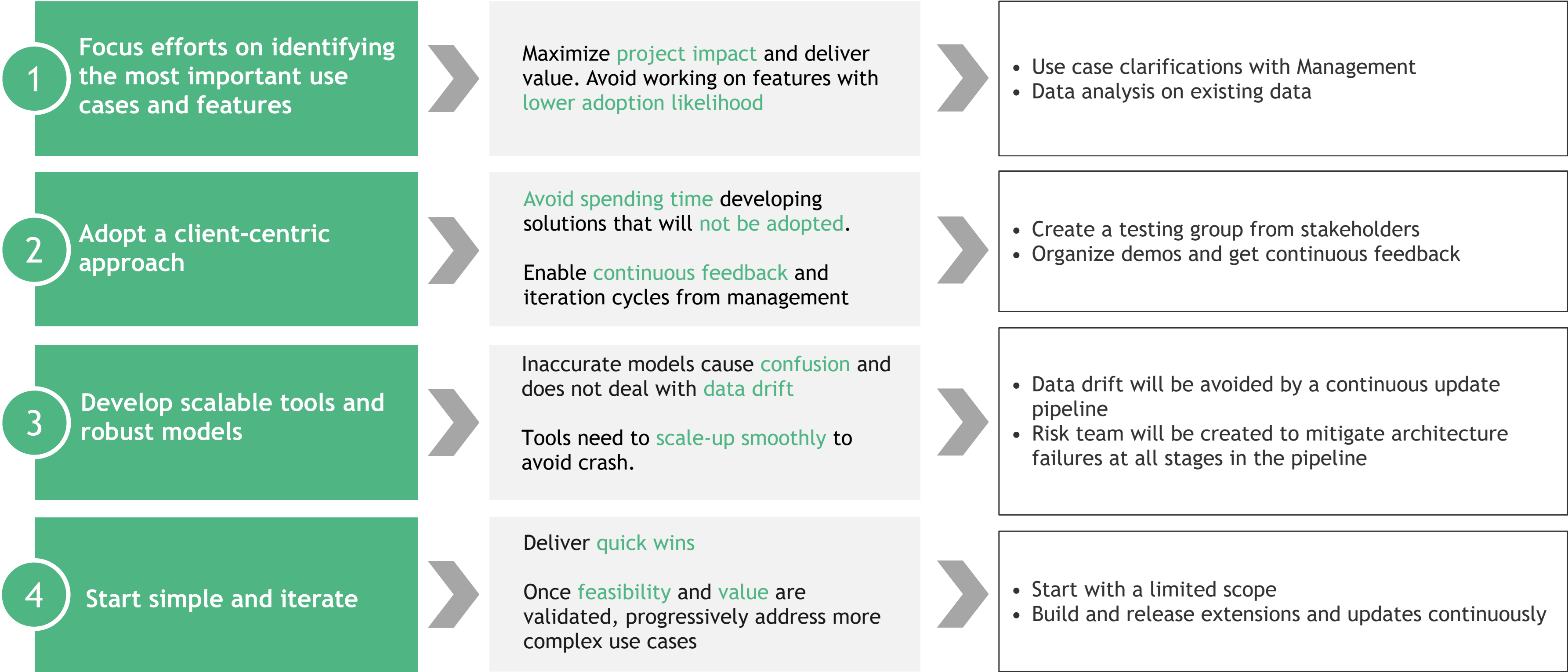


Key Success Factors


Key Success Factors


Why?

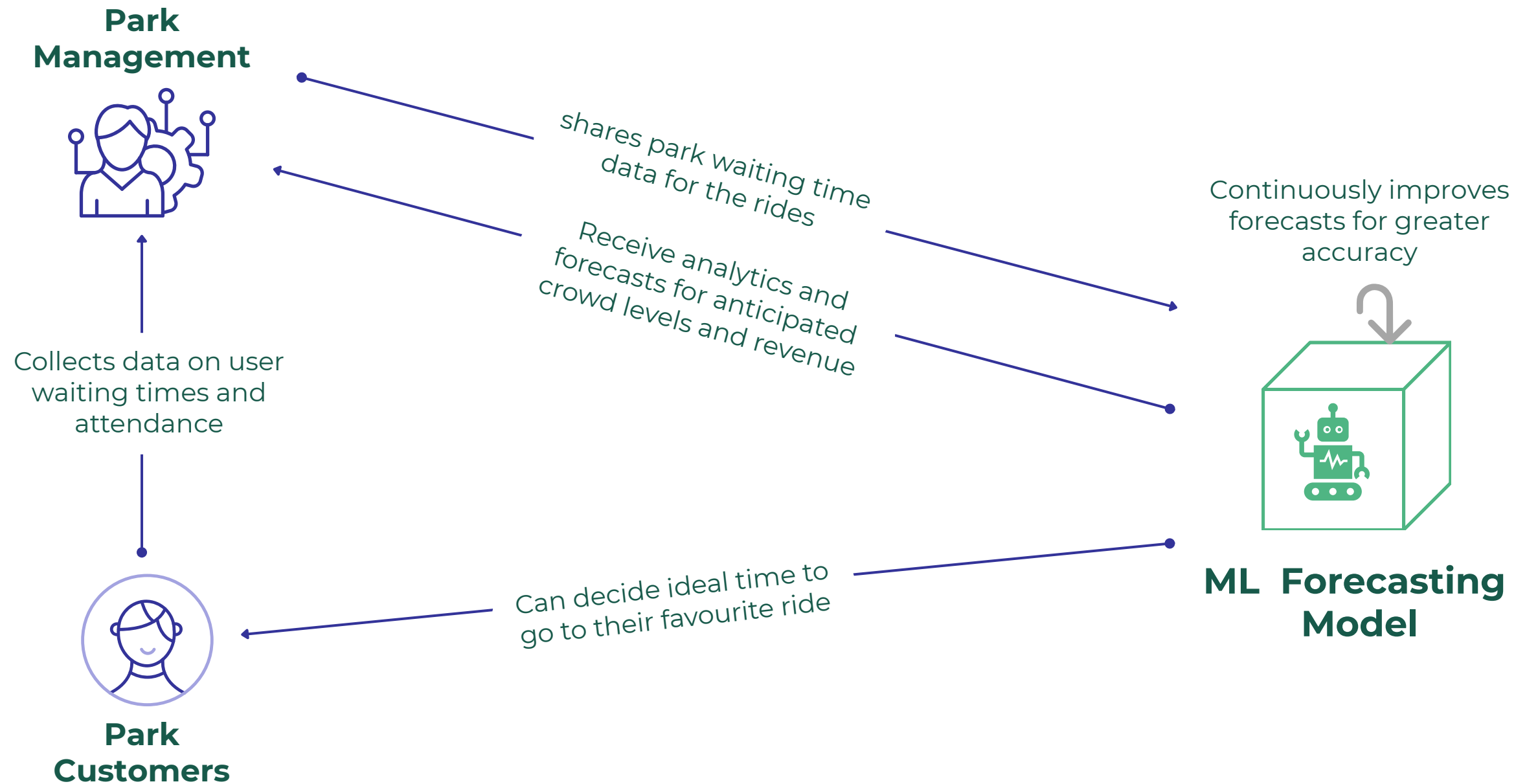
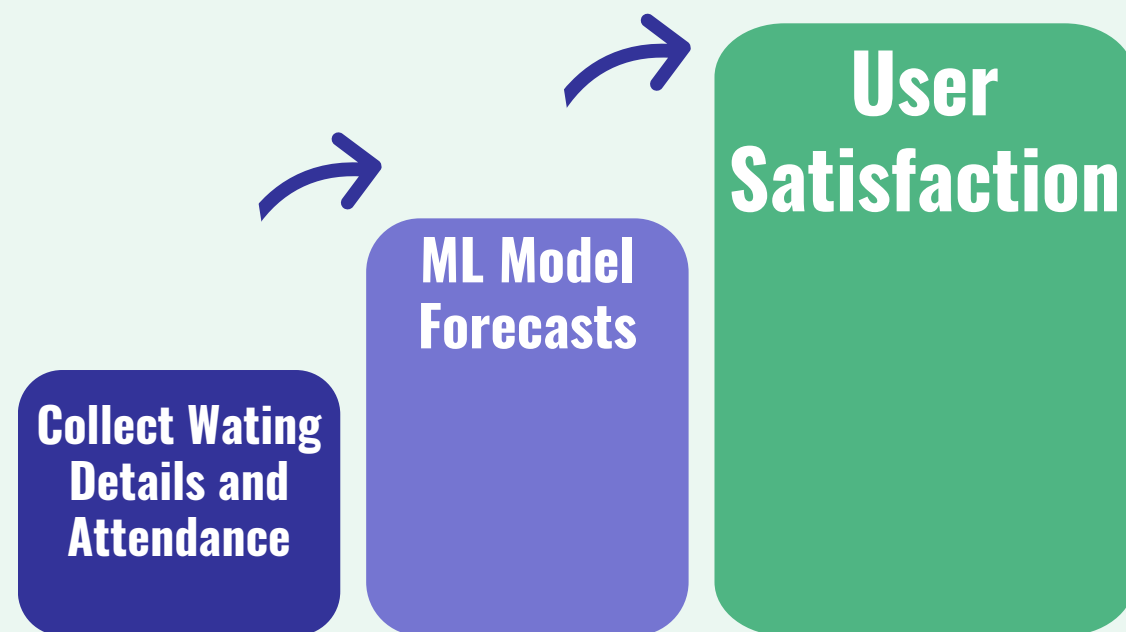

How?



Key Technical Concepts

From Descriptive → Predictive → Prescriptive

Benefits of switching to our ML model for predictions

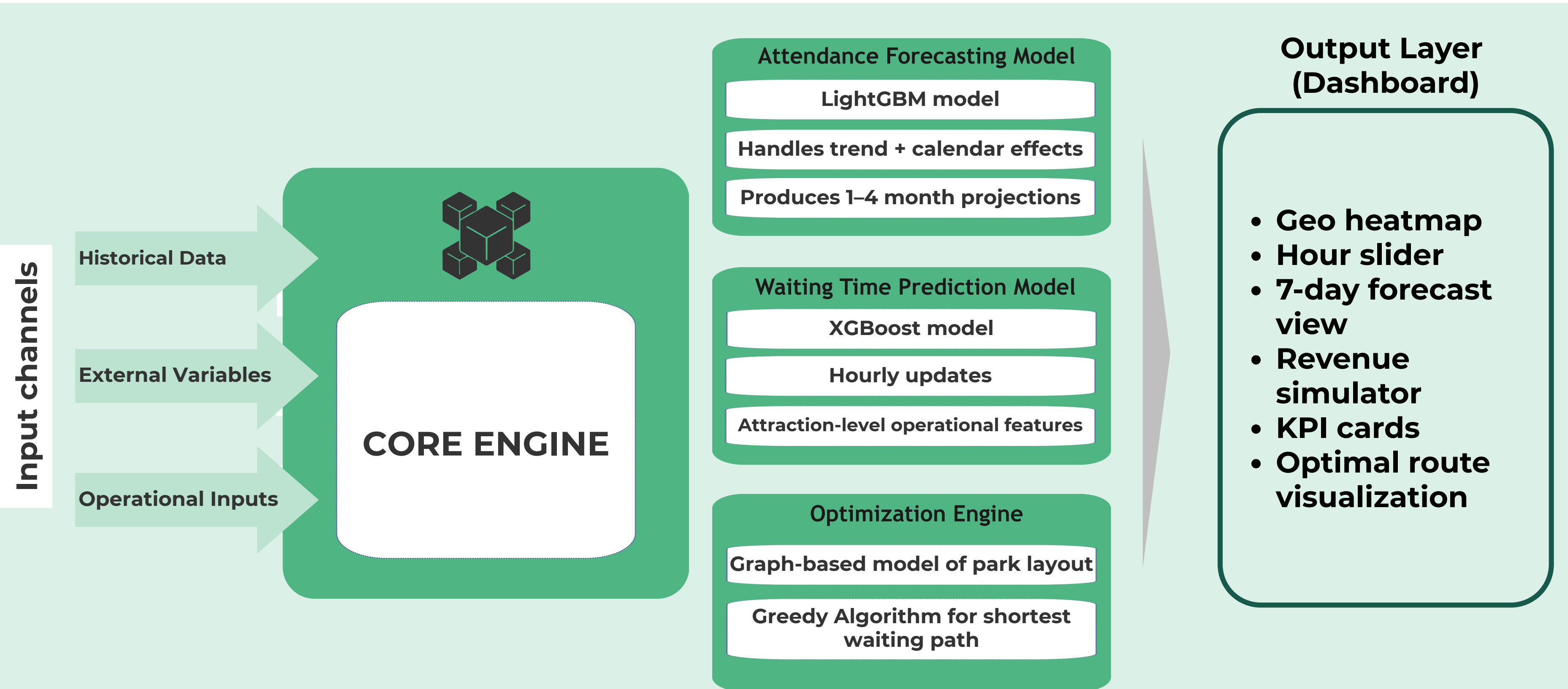


- For Park Customers (Visitors)**
- Smarter visit planning – Know the ideal time to go to favorite rides
 - Reduced wait times – Avoid long queues with crowd-informed timing
 - Better experience – Less guessing, more enjoying

- For Park Management**
- Data-driven decisions – Receive analytics and forecasts for crowd levels and revenue
 - Operational efficiency – Anticipate demand and allocate resources accordingly
 - Improved communication – Share accurate wait time data with guests proactively

System Architecture

From Descriptive → Predictive → Prescriptive



Scope

Defining the Scope of the Project

POC Scope

Forecasting

- Hourly waiting time forecasts (1 week horizon)
- Attendance forecast (up to 4 months)
- Revenue simulation

Dashboard

- Interactive geo heatmap
- Time slider (hour-by-hour)
- Attendance & revenue projection view
- Optimal route planning demo

Full Scope

- Real-time IoT crowd tracking WebApp
- Behavioral modeling of individual visitors
- Full ERP integration
- Staffing optimization engine

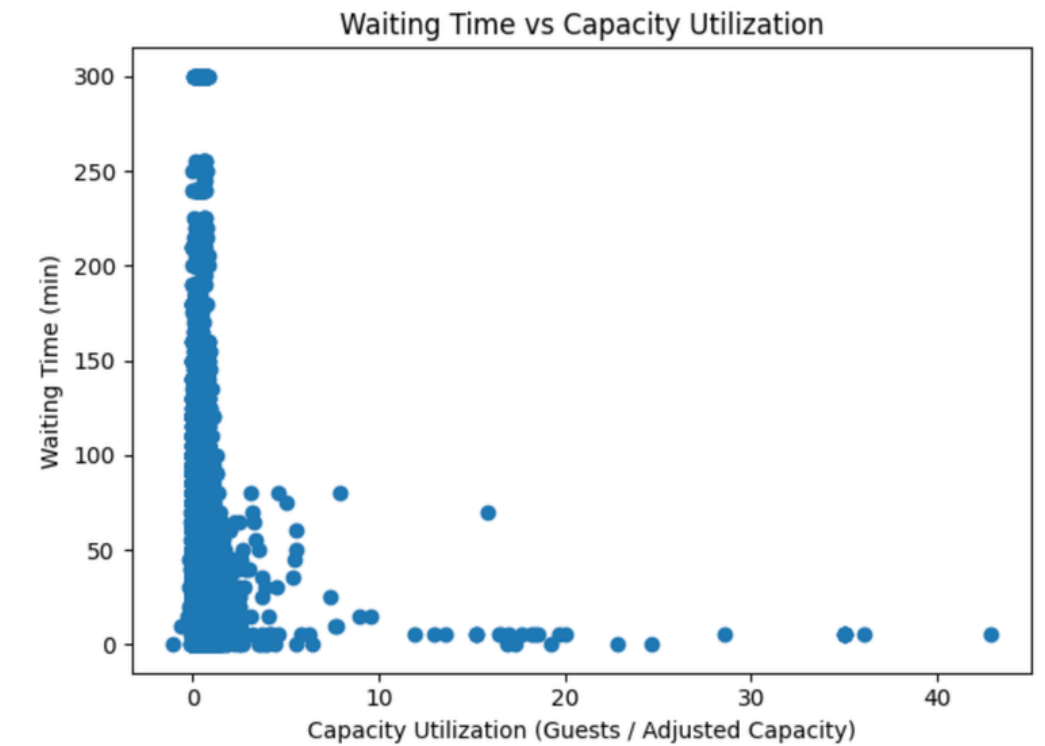
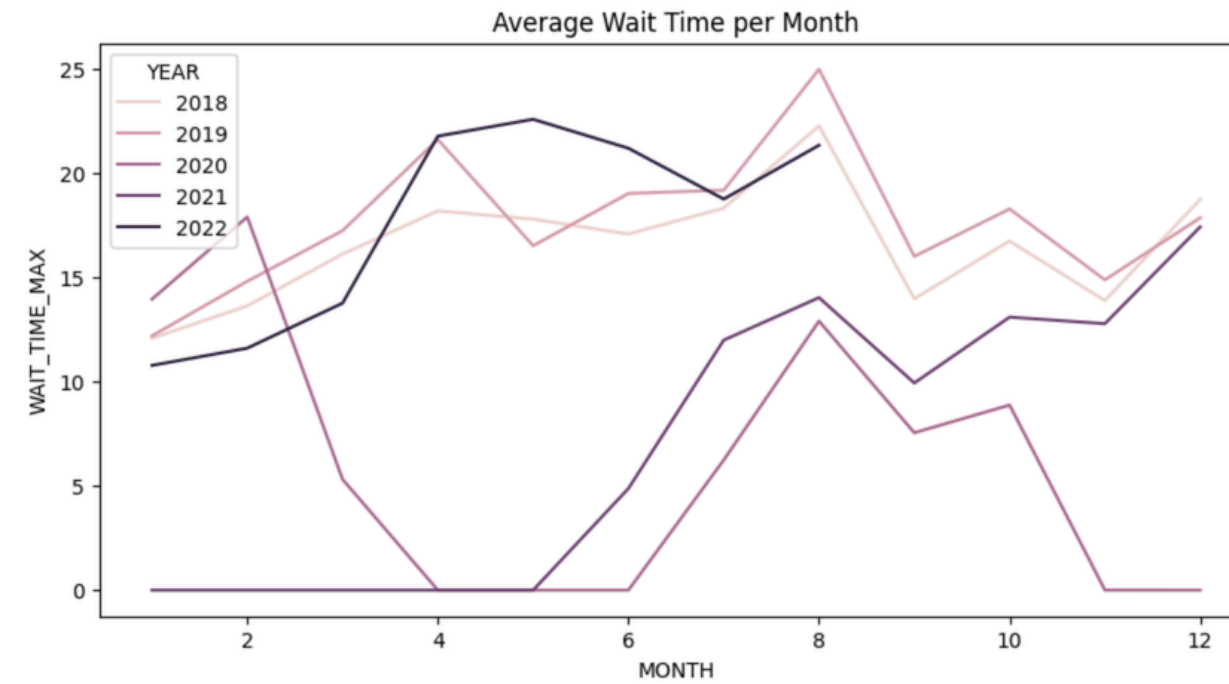
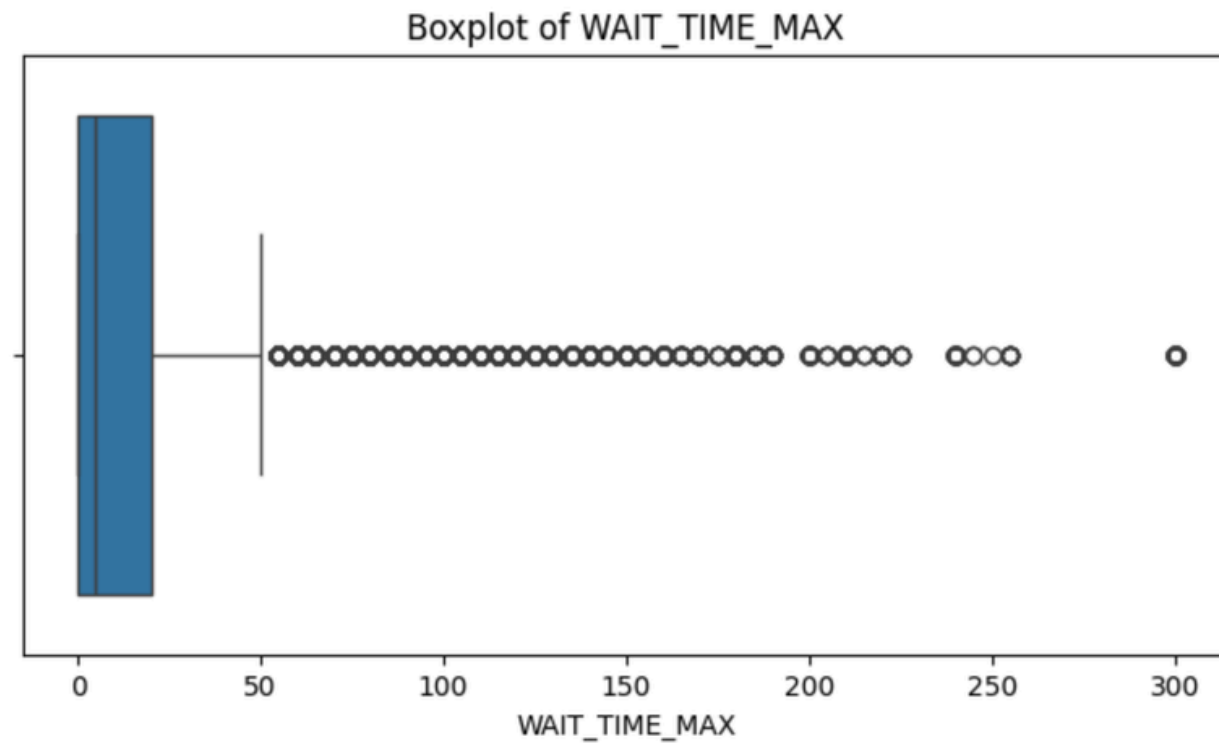


We move from predicting congestion to actively optimizing the park's operational and commercial performance.

Key Insights (EDA)

Waiting_time_csv

- For the EDA we followed the checklist proposed by Sania Panchal in her *Medium* article: [EDA in 30 Minutes - My Checklist for Quick Understanding](#).



Wait Time

- Highly left-skewed
- Median $\approx$ low, but long tail up to extreme values (12')
- Most waits are short, but congestion events exist (outliers)
- Very high variation (19')

Time Evolution (graph 2)

- Clear structural break around 2020 (Covid)
- Drop in occupancy and wait times
- Recovery trend after 2021
- Seasonality trends

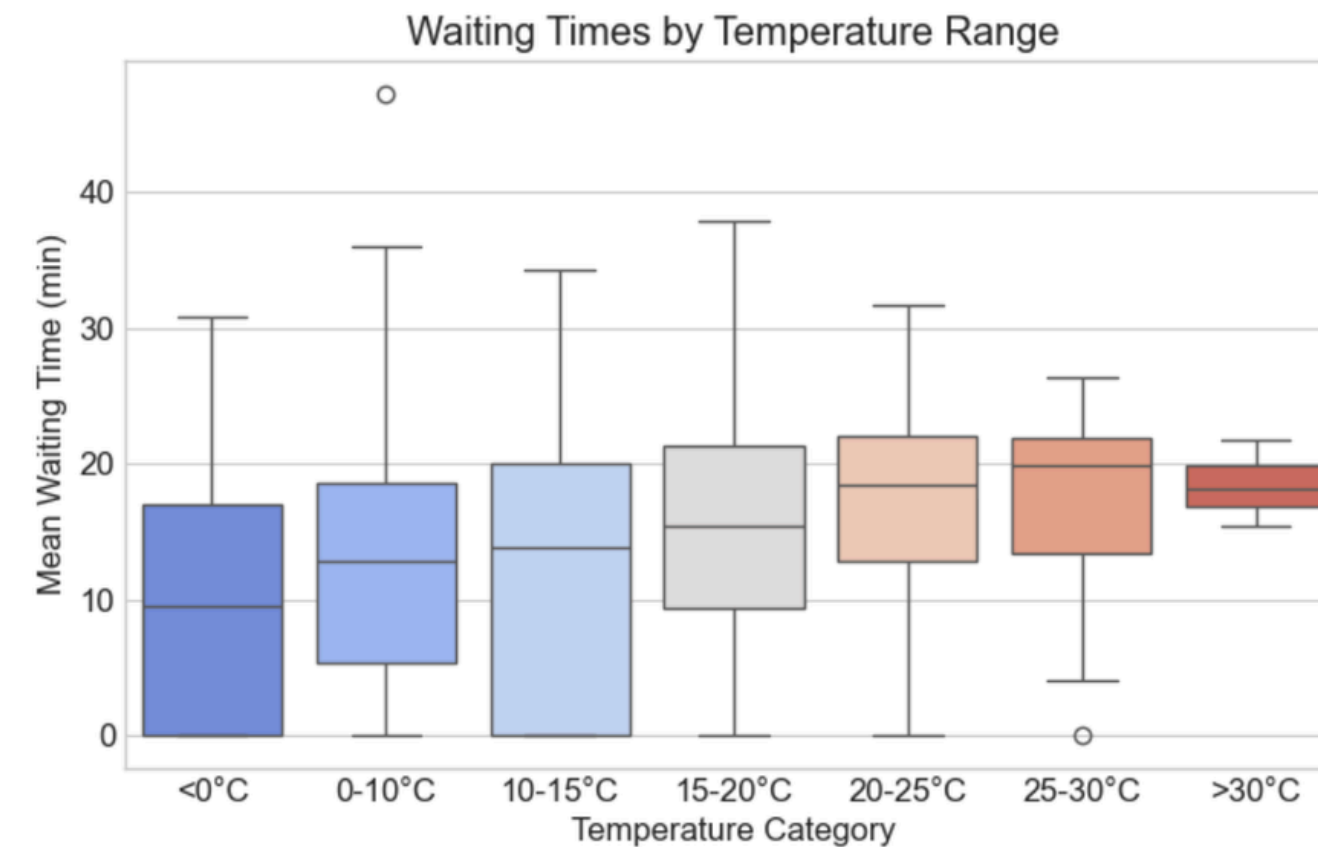
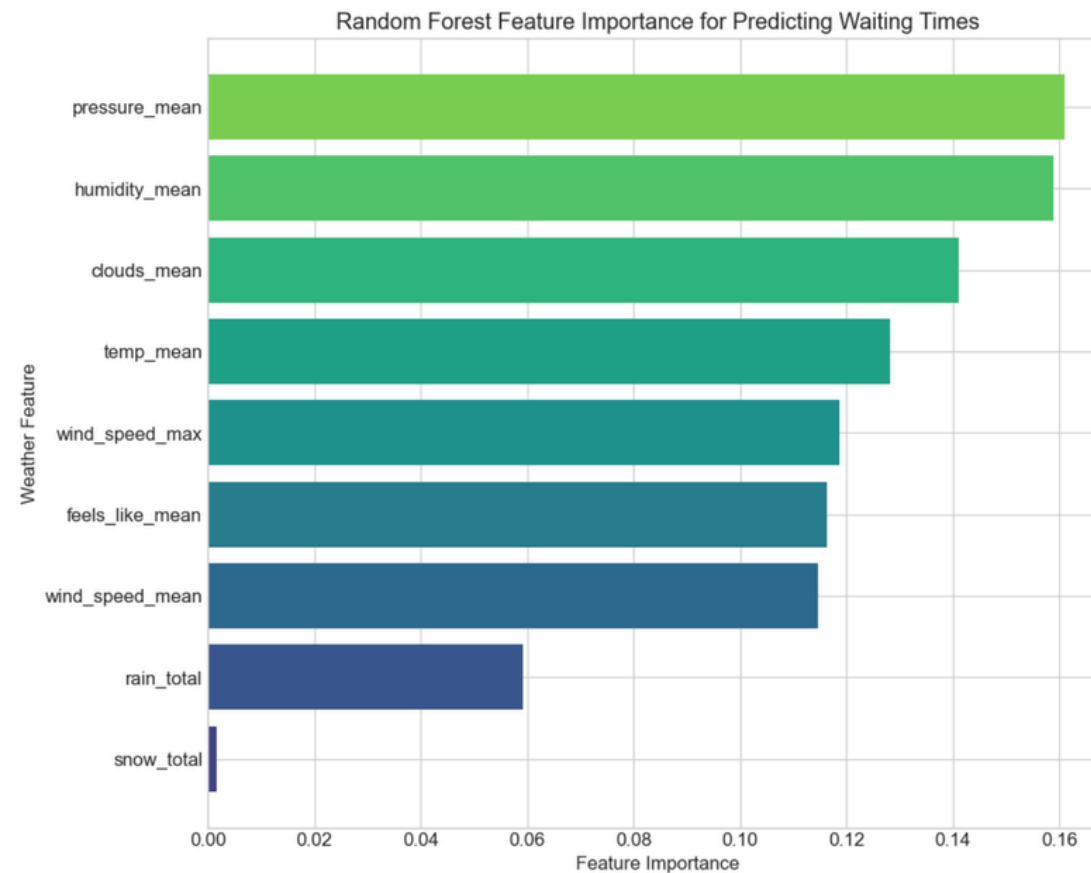
Occupancy vs Wait Time (graph 3)

- Weak / noisy relationship between occupancy and waiting time
- High dispersion: similar occupancy $\rightarrow$ very different wait times
- Occupancy alone does not explain congestion

Key Insights (EDA)

weather_data_csv

- For the EDA we followed the checklist proposed by Sania Panchal in her *Medium* article: [EDA in 30 Minutes - My Checklist for Quick Understanding](#).



Waiting time:

- Composite weather conditions matter more than temperature alone
- Pressure, humidity, and cloud cover rank highest
- Temperature adds signal but doesn't dominate
- Rain explains rare low-wait outliers
- Weather = secondary driver (calendar/ops dominate)

Waiting Time vs Temperature (Boxplot)

- Waiting times increase with temperature (monotonic trend)
- Cold days → lower median waits
- Warm days (>20°C) → higher median waits
- Higher dispersion in mid-range temperatures
- Heat amplifies congestion risk

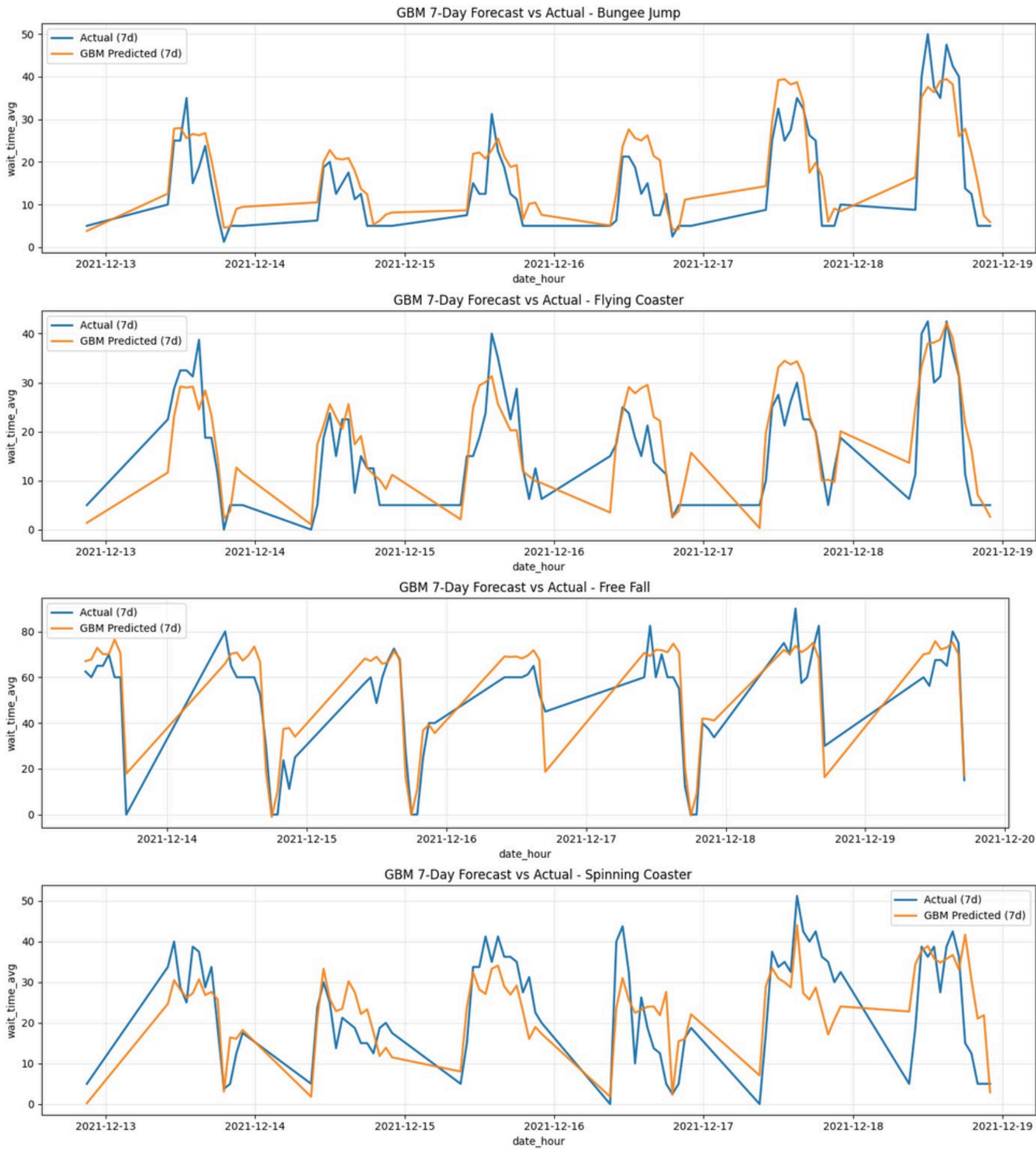
Forecast Results

7-Day Hourly Waiting Time Forecast – Model Performance

Key Takeaways

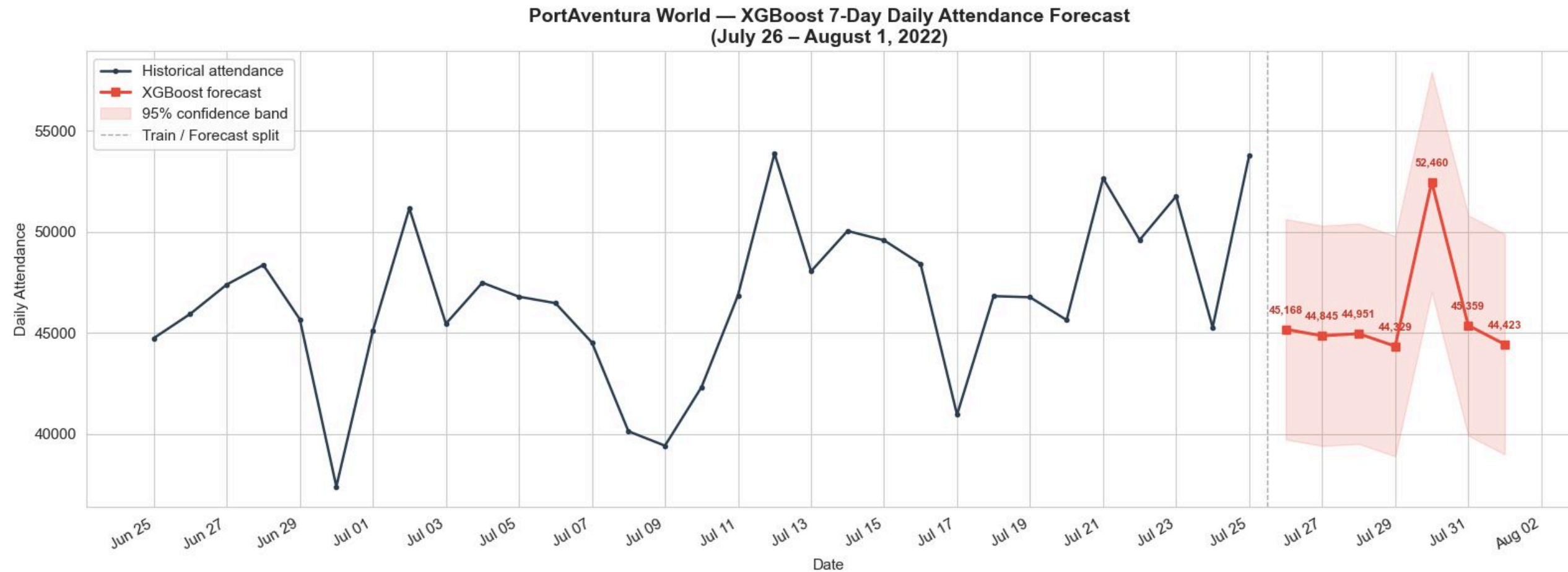
- Model predicts hourly waiting times with 5 - 9 minutes average error
- Accuracy is strongest for medium-volatility attractions (≈5 min MAE).
- Higher MAE (Free Fall – 8.9 min) reflects high peak variability and sharp drops, which are harder to anticipate
- Forecasts capture daily patterns, peaks, and operational resets reliably

Attraction	MAE (minutes)
Bungee Jump	5.5
Flying Coaster	5.3
Free Fall	8.9
Spinning Coaster	7.5



Forecast Results

7-Day Attendance Forecast – Model Performance & Insights

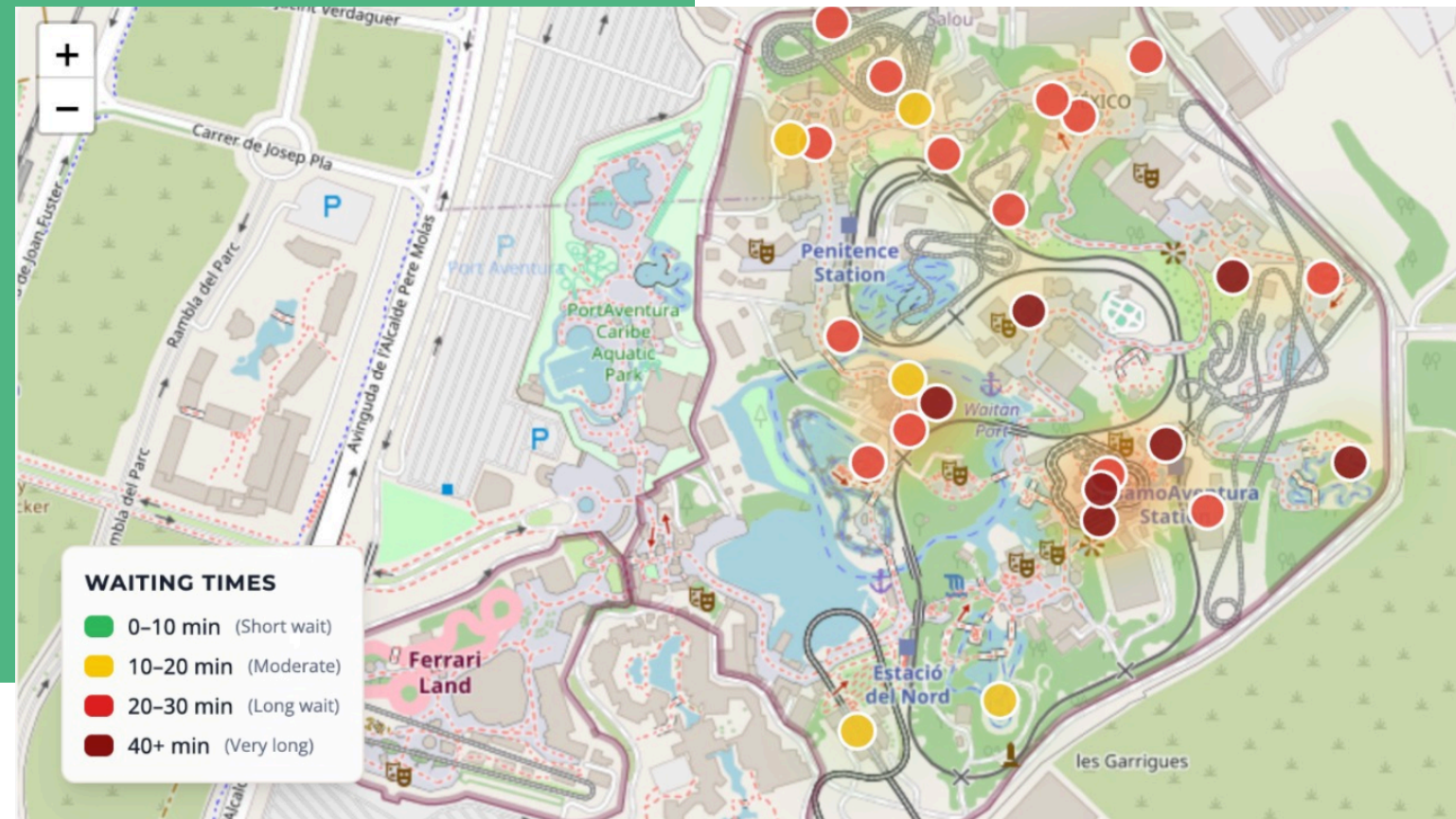


What the Model tell us:

- The model captures trend + weekly effects well
- The forecast continues this pattern with:
 - ~45k visitors on weekdays
 - A strong weekend peak (~52k on July 30)
- Uncertainty is moderate and acceptable for operational planning

Why both forecasts matters:

- Provides actionable short-term visibility, enabling:
 - Proactive staffing decisions
 - Better queue management
 - Revenue optimization during peak days
 - Reduced operational surprises



OBJECTIVE 1

DASHBOARD: OPERATIONAL VIEW

This provides a 7-day forecast of waiting times at attraction level, visualized through an interactive geo-heatmap of the park.

It enables operational teams to anticipate congestion, optimize visitor flow, and reduce average waiting times through proactive decision-making.



7-Day Waiting Time Prediction

- Hour-by-hour predictions at attraction level
- Capture effects of attendance, weather, and events
- Quantify waiting time ranges & detect peak congestion windows before queues form

Provides forward visibility for operational planning



Interactive Park Heatmap

- Visual congestion intensity across the park
- Color-coded waiting time clusters
- Time slider to simulate day & hour
- Identify spatial crowd imbalances

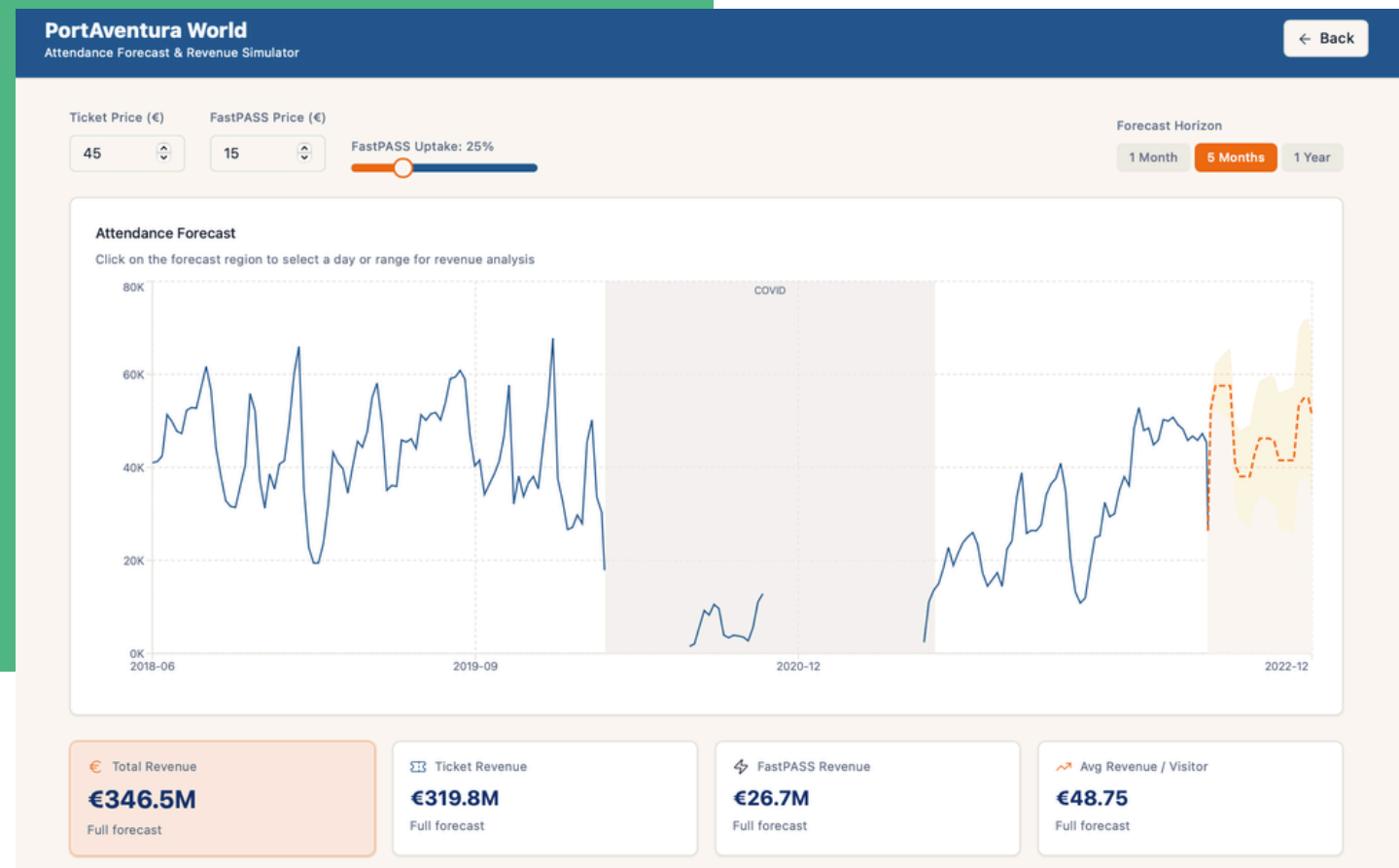
Enables intuitive understanding of how visitors redistribute across the park



Dynamic Visitor Flow Optimization

- Compute least-waiting-time path between attractions
- Quantify potential reduction in total waiting time

Demonstrates measurable operational impact.



OBJECTIVE 2

DASHBOARD: STRATEGIC VIEW

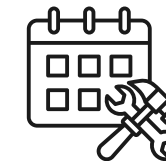
A strategic planning dashboard leveraging medium-term attendance forecasts to anticipate demand peaks, optimize resource allocation, and support revenue-driven decisions.



Seasonality & Demand Anticipation

- Daily attendance projections over 1, 5 and 12 Months.
- Capture seasonal patterns and peak periods
- Identify high-demand weekends & holidays
- Scenario simulation (base / high-demand cases)

Provides visibility for capacity & staffing planning



Resource & Maintenance Optimization

- Anticipate staffing needs ahead of peaks
- Enables scheduling rehabilitation during low-demand windows
- Adjust ride availability strategy
- Support long-term planning

Reduces operational stress during peak attendance



Strategic Revenue Levers

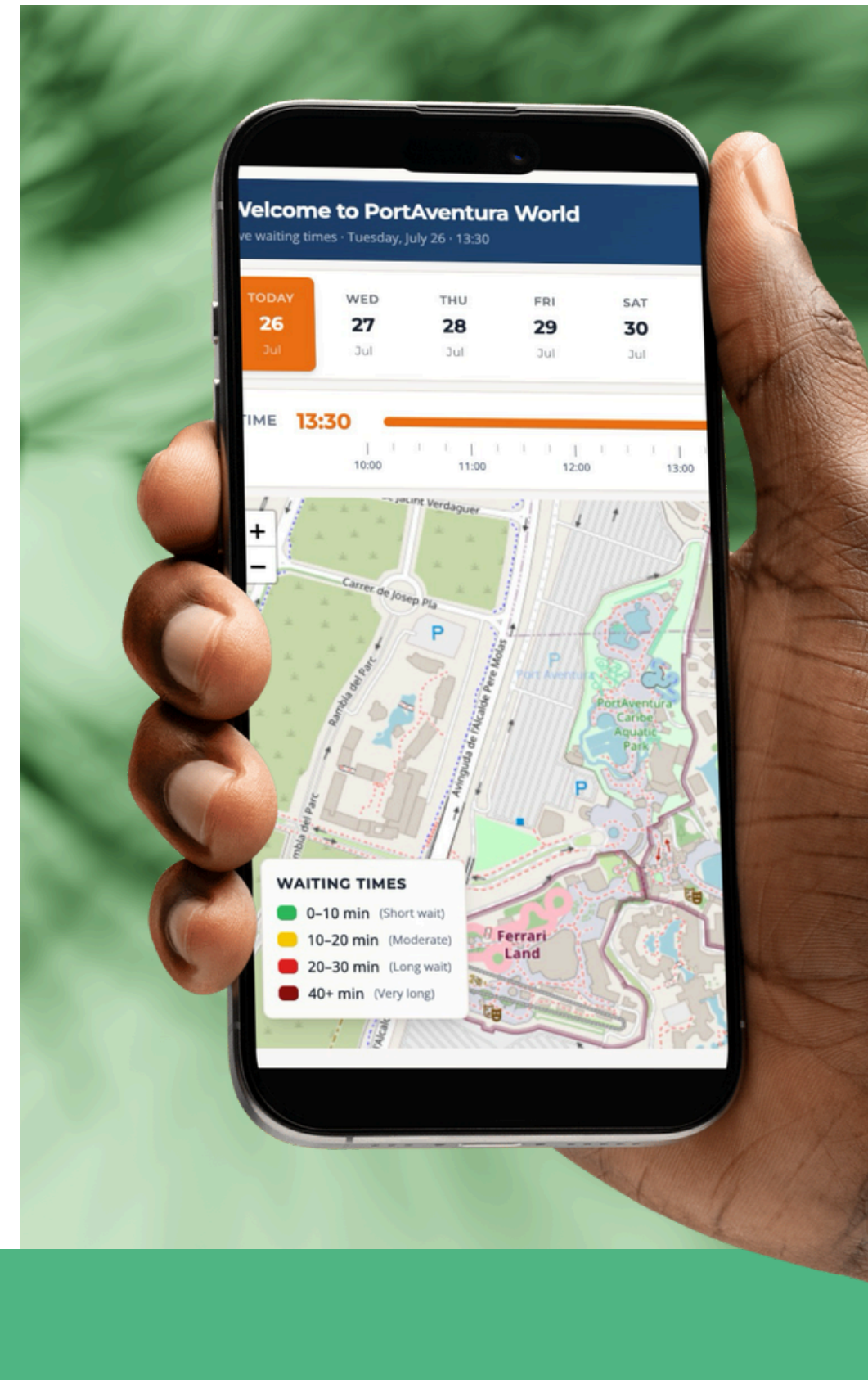
- Fastway pricing sensitivity during peaks
- Estimate revenue impact of attendance
- Simulate revenue scenarios under different demand assumptions

Translates demand forecasting into financial decision support

DEMO

HEATMAP & CROWD ROUTING

TRY THE LIVE DEMO !



VALUE CREATION

COMMERCIAL IMPACT

↓ Peak Waiting Time Reduction

Through proactive routing, staffing adjustments and congestion anticipation

- Faster queue turnover
- Reduced ride downtime stress
- Better resource allocation

Visitor Satisfaction Uplift

Reducing unpredictability and improving crowd flow

- Higher NPS
- Increased repeat visits
- Stronger brand perception

Fastway & Upsell Revenue Potential

Revenue growth without infrastructure expansion

- Dynamic pricing & capacity optimization
- Optimized Fastway pricing during peaks



Timeline

Upgrade MVP

- Clean and modularize forecasting engine
- Improve heatmap accuracy & hourly refresh
- Refine optimal route logic (real waiting weights)
- UX redesign for customer-facing interface

Month 1

Customer Web App Deployment

- Deploy public web interface (mobile-friendly)
- Live geo-heatmap with hour slider with a 7-day forecast view and optimal route visualization for visitors

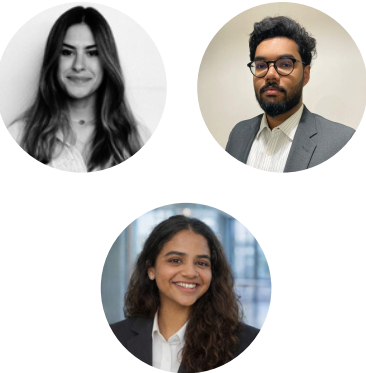
Month 3

Month 2

Management Layer & Enhancements

- Add revenue simulator panel
- Add KPI cards (forecast vs actual, congestion index)
- Admin controls for manual overrides
- Performance optimization & load testing

Why Us?

		Role	Dedicated time	Responsibilities
Technical Leads		Forecasting and ML expertise Strategic and technical expertise	20 hours a week	R A C I
Project Lead		Strategic framing Coordination Operational follow-up of deliverables	20 hours a week	R A C I
Consultants		Dashboard Creation Weekly meetings Insights Deployment of Dashboard	20 hours a week	R A C I

R

 Responsible

A

 Accountable

C

 Consulted

I

 Informed

LEADERSHIP TEAM

Get to know the team that will be taking on the project



Alessandro Ivashkevich

Data Scientist

Built interactive map visualization of forecasted waiting times. Integrated attendance and weather forecasting into the models



Artur Garipov

Data Scientist

Developed hourly waiting time forecasting models (SARIMAX, XGBoost), focusing on accuracy, feature engineering, and model validation. Developed data pipeline.



Joshua Ebenezer

Product Lead

Developer turned product lead with over 3 years of experience in cool tech SaaS products. Cool head around tight deadlines prioritizing key metrics



Mathilde Lafon

Business & Dashboard

Design the operational heatmap interface and strategic revenue simulator, ensuring intuitive UX and actionable insights



Navneeth Rajesh

Commerical Strategy

Quantify value creation (pricing optimization, FastPASS penetration, staffing impact) and structured ROI narrative for management.



Sarah George

Product & Implementation

Defining roadmap from demo to production WebApp, including ERP integration vision, IoT extensions, and long-term scalability



THANK YOU



BLOOPERS

We had a really good time hustling out the models and UI



APPENDIX

OLS Regression Results

```
=====
Dep. Variable:          wait_time_avg    R-squared:                0.812
Model:                  OLS              Adj. R-squared:           0.811
Method:                 Least Squares    F-statistic:             2.298e+06
Date:                  Tue, 10 Feb 2026  Prob (F-statistic):      1.00e-70
Time:                  13:33:10          Log-Likelihood:          -81483.
No. Observations:      23283            AIC:                    1.632e+05
Df Residuals:          23189            BIC:                    1.639e+05
Df Model:               93
Covariance Type:       cluster
=====
```

	coef	std err	z	P> z	[0.025	0.975]
covid	18.2101	4.368	4.169	0.000	9.648	26.772
post_covid	-4.5820	5.345	-0.857	0.391	-15.058	5.894
utilization:covid	-33.6637	8.097	-4.158	0.000	-49.534	-17.794
utilization:post_covid	11.2931	8.745	1.291	0.197	-5.847	28.433